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HORNBILL: A First-in-Human Phase I/IIa Study of the Safety, Tolerability, and Early Pharmacodynamics of BI 764524 for Diabetic Macular Ischemia

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Objective: To report safety and early pharmacodynamic results from a first-in-human trial of intravitreal (IVT) anti-semaphorin 3A antibody in participants with diabetic macular ischemia (DMI).

Design: HORNBILL, a phase I/IIa study of BI 764524, comprised a nonrandomized, open-label, uncontrolled, single-rising-dose (SRD) and masked, randomized, sham-controlled, multiple-dose (MD) parts.

Participants: Adults with DMI and stable diabetic retinopathy (DR) treated with pan-retinal photocoagulation and without center-involving diabetic macular edema.

Methods: Twelve participants received single IVT doses of BI 764524 0.5 mg (n = 3), 1.0 mg (n = 3), or 2.5 mg (n = 6) in the SRD part. Thirty-one participants received 3 IVT doses of BI 764524 2.5 mg (n = 21) or sham procedures (n = 10) at 4-week intervals and were followed to week 22 in the MD part.

Main Outcome Measures: The primary SRD end point was the number of participants with dose-limiting events; secondary end points assessed drug-related and ocular adverse events (AEs). The primary MD end point was the number of participants with drug-related AEs; secondary end points included changes from baseline in foveal avascular zone (FAZ) area, best-corrected visual acuity (BCVA), and central subfield thickness (CST).

Results: No dose-limiting events or drug-related AEs were reported with SRD; the highest tested dose (2.5 mg) was selected for the MD part. In the MD part, 2 investigator-assessed drug-related AEs (vitreous floaters and increased gamma-glutamyl transferase) were reported. No intraocular inflammation or occlusive retinal vasculitis cases occurred. At week 12 (4 weeks after the final injection), the adjusted mean FAZ area change was −0.004 mm^2 in the BI 764524 group and +0.019 mm^2 with sham. At week 22 (14 weeks after the final injection), the adjusted mean FAZ area change was −0.001 mm^2 in the BI 764524 group and +0.010 mm^2 with sham. No relevant BCVA and CST changes occurred.

Conclusions: HORNBILL met the primary safety end points; all evaluated BI 764524 doses were well tolerated. These findings support further investigation of BI 764524 in participants with DR and retinal nonperfusion.

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Diabetic retinopathy (DR) is a leading cause of preventable vision loss in working-age adults.[1,2] Retinal nonperfusion (RNP) is a hallmark of DR, driving its development and progression.[3–5] Persistent RNP in the central retina, accompanied by damage to the macular vasculature, can lead to diabetic macular ischemia (DMI).[4–6] There are currently no efficacious and approved treatments preventing the progression of RNP secondary to DR or slowing the onset and progression of DMI.[7,8]

Data on the prevalence of DMI in patients with DR are limited, and available estimates vary according to different definitions and imaging modalities. To date, DMI is reportedly most prevalent in patients with advanced disease (i.e., proliferative DR).[6,9] Diabetic macular ischemia is characterized by irregular enlargement and disruption in the foveal avascular zone (FAZ) and progressive macular capillary loss.[9] Over time, this macular capillary loss can lead to neuronal cell death, which, in some cases, renders the vision loss irreversible.[9–11]

Although imaging and measurement of macular capillary loss has improved substantially since the introduction of OCT angiography (OCTA) imaging,[3,8] DMI research

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remains challenging due to a lack of consensus regarding imaging biomarkers of RNP.[8] Some proposed imaging biomarkers include FAZ area,[9] macular vascular density,[12] and geometric perfusion deficits.[13] Foveal avascular zone area has been suggested to be the best-studied biomarker of central RNP on OCTA.[11]

Semaphorins are a broad family of signaling proteins that are pivotal in directing axonal growth and vascular development.[14] Semaphorin 3A (Sema3A) binds to the co-receptor neuropilin-1 to induce vasorepulsive effects and increase retinal vascular permeability.[15–17] Secreted by retinal ganglion cells under hypoxic conditions, Sema3A forms a chemical barrier that repels new blood vessels away from the ischemic retina and toward the vitreous.[15,16] In addition, raised Sema3A levels have been found in the vitreous and plasma of patients with proliferative DR and those with diabetic macular edema.[18,19] As such, blocking Sema3A may help to reduce retinal ischemia or vascular permeability, or both, found in patients with DR.

BI 764524 is a novel agent that is administered intravitreally to bind and neutralize Sema3A and is under investigation in DMI. HORNBILL (A First-in **H**uman Trial to Study Safety and Tolerability of Single Rising Intravitreal d**O**ses [open label, nonrandomized, uncontrolled] and in Addition the Early Biological Response of Multiple intravit**R**eal Dosing [single-masked, ra**N**domized, sham-controlled] of **BI** 764524 in panretina**L** photocoagulation-treated pro**L**iferative diabetic retinopathy patients with diabetic macular ischemia; NCT04424290) was a 2-part phase I/IIa study that examined the safety and early pharmacodynamics of intravitreal (IVT) BI 764524 in patients with DMI secondary to DR. Herein, we present results from this first-in-human study.

Methods

HORNBILL comprised a nonrandomized, open-label, uncontrolled, single-rising-dose (SRD) part and a randomized, masked, sham-controlled, multiple-dose (MD) part (Fig 1) and investigated the safety, tolerability, and early pharmacodynamic activity of BI 764524 in adults with DMI. The HORNBILL study was conducted across 19 clinical centers in the United Kingdom and the United States. Institutional review board/ethics committee approval was obtained. A complete list of study centers is available at ClinicalTrials.gov (NCT04424290).

Participants

Eligibility Criteria. Eligible participants were aged ≥18 years with the presence of DMI and had stable DR after previous pan-retinal photocoagulation treatment for pro**l**iferative DR. The required hemoglobin A1c level was ≤12.0%. The study was conducted in accordance with the principles of the Declaration of Helsinki and the International Conference on Harmonisation Good Clinical Practice guidelines. All study participants provided written informed consent before enrollment.

For the SRD part, eligible participants had DMI as evidenced by macular capillary loss, defined as disruption of retinal vascularity in superficial or deep-retinal plexus, or both, on OCTA, as determined by the investigator, and best-corrected visual acuity (BCVA) ≤55 letters (the BCVA of the nonstudy eye had to be better than or equal to the BCVA of the study eye).

For the MD part, eligible participants had significant DMI, defined as an enlarged FAZ (≥0.5 mm^2 area in the superficial vascular complex) on OCTA, or if the FAZ was <0.5 mm^2, an enlarged perifoveal intercapillary space in ≥1 quadrant was deemed acceptable for inclusion, as assessed by a centralized reading center. Best-corrected visual acuity ≤85 letters was also required, and either eye could be chosen by the investigator as the study eye if both were eligible.

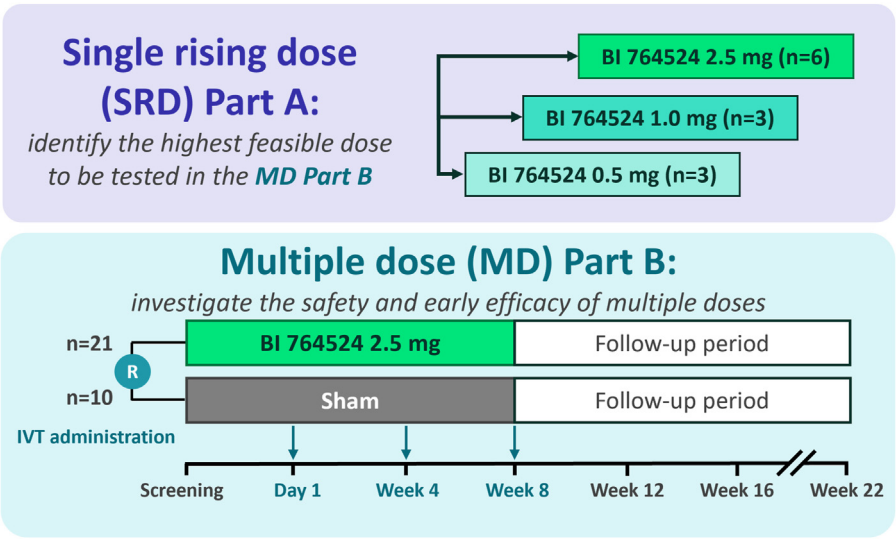


Figure 1. Design of the SRD and MD parts of the HORNBILL study. In part A, dose escalation was decided by a safety monitoring committee and was based on a review of safety data of the previous dose and guided by a Bayesian logistic regression model-based approach with overdose control. In part B, vertical blue arrows indicate the timing and frequency of IVT injections during the treatment period. IVT = intravitreal; MD = multiple dose; R = randomization; SRD = single rising dose.

Participants were excluded if they had diabetic macular edema (MD part only); received IVT anti-VEGF injections, steroids, or treatment with a macular laser within 3 months of enrollment; had a condition in the study eye other than DR that might affect BCVA or the interpretation of results (e.g., epiretinal membrane or heavily lasered macula); had intraocular surgery within 3 months of screening; had active intraocular inflammation in the study eye; had active infectious conjunctivitis in either eye; or had anterior segment or vitreous abnormalities that would preclude adequate imaging. If both eyes were eligible for inclusion and had identical visual acuity, the study eye was selected by the investigator. Full details of selection inclusion and exclusion criteria are provided in Table S1 (available at www.ophthalmologyscience.org).

Dosing

SRD Part. Participants received single IVT doses of BI 764524 (0.5, 1.0, or 2.5 mg). Treatment was administered at the first visit after the screening period, with a follow-up of 14 weeks. In each dose cohort, a single patient was treated on day 1; if this patient did not experience an ocular or systemic serious adverse event (SAE) or any clinically significant abnormality by day 4, the remaining patients were dosed (Fig 1). All patients within 1 dose cohort were observed for a minimum of 7 days after study drug administration, including the assessment for dose-limiting events before the decision for dose escalation was made. The dose escalation was decided by a safety monitoring committee, based on a review of safety data of the previous dose cohort and guided by a Bayesian logistic regression model-based approach with overdose control. The safety monitoring committee provided a recommendation on the dose selection for the MD part.

MD Part. Each participant in the MD part received IVT BI 764524 2.5 mg or sham injection procedures (randomization ratio 2:1) every 4 weeks over an 8-week period (overall 3 injections), with an additional 14-week follow-up (total MD part duration, 22 weeks; Fig 1). Treatment assignment was masked to all study center personnel and participants. The IVT injections or sham injection procedures were administered by an unmasked injector that had no other involvement in the trial.

End Points

SRD Part. The primary end point in the SRD part was the number of participants with dose-limiting events from drug administration to day 8 (7 days posttreatment). Secondary end points were the number of participants with drug-related adverse events (AEs) and the number of participants with ocular AEs, each from drug administration until the end of study. Further end points included change from baseline in FAZ area, BCVA, and central subfield thickness (CST) at each visit.

MD Part. The primary end point in the MD part was the number of participants with drug-related AEs from drug administration until the end of study (week 22). The secondary safety end point was the number of participants with ocular AEs until the end of study. Additional secondary end points were changes from baseline in FAZ area at week 12 (visit 5) and at week 22 (visit 7, assessed by OCTA) and changes from baseline in BCVA and CST (assessed by spectral-domain OCT) at visits 3−7. Safety and early pharmacodynamic responses were examined up to 14 weeks after the last injection.

OCTA Image Acquisition and Analysis

Retinal images were acquired using spectral-domain OCTA (Avanti RTVue-XR, Optovue). Foveal avascular zone metrics were evaluated on the inner retina slab, referred to afterward as full-thickness retina (FTR), bounded by the internal limiting membrane and the outer plexiform layer and based on 3 × 3-mm OCTA scans. A single masked grader performed grading at an independent and masked reading center (Ocular Imaging Research and Reading Center). Foveal avascular zone boundaries were considered undetermined if they were continuous with largely ischemic regions that extend outside the 3 × 3-mm field.

Before any FAZ measurements were reported, the segmentation of the FAZ slab boundaries, as defined, was checked and verified by the reading center. Manual adjustment of the segmentation was made if necessary. A value was reported only if the segmentation was considered reliable: the scan quality index was ≥6, and no moderate or severe artifacts were observed within the fields where the FAZ was present. The FAZ boundaries were manually delineated by a single masked grader utilizing 3 × 3 mm FTR scans. Any queries with the grading forms for certain review cases were handled by a senior masked grader (M.A.). A detailed description of the OCTA image acquisition protocol is provided in the Supplementary Methods (available at www.ophthalmologyscience.org).

Additionally, manual unmasked inspection of the OCTA images collected during HORNBILL was performed to investigate the effect of BI 764524 on RNP, which is subjective by nature. Further information on OCTA acquisition and analysis can be found in the Supplementary Methods.

Spectral-Domain OCT Image Acquisition and Analysis

Spectral-domain OCT (Spectralis spectral-domain OCT, Heidelberg Engineering, Inc) scans were also used to monitor the retina. Forty-nine B-scan volume raster scans were obtained. The segmentation of each B-scan was checked and adjusted for internal limiting membrane and Bruch's membrane. The ETDRS grid was centered to the fovea. Central subfield thickness (μm) was measured using in-built software (Heyex, Heidelberg Engineering). Automated segmentation lines were manually checked for accuracy, and manual adjustments were done wherever necessary before the readings were considered final.

Statistical Analyses

The treated set included all patients randomized and treated with ≥1 dose of study drug (BI 764524) or receiving ≥1 sham injection. Primary and secondary end points were analyzed using the treated set. No testing of statistical hypotheses was planned. Specific analyses pertaining to the SRD and MD parts are described next.

SRD Part. No formal statistical power calculations to determine sample size were performed. Dose escalation was guided by a Bayesian logistic regression model-based approach, based on a weekly informative prior distribution, and employing the escalation with overdose control principle.

MD Part. The minimal sample size (N = 30 [n = 20, BI 764524; n = 10, sham]) was calculated based on the estimated likelihood of observing prespecified treatment effects for different scenarios, given the previously derived mean intraindividual variability of FAZ estimates.[20] Data over time were analyzed using a mixed model with repeated measures, with the baseline value as a continuous covariate; visit and treatment, and visit−treatment and visit−baseline interactions as categorical fixed effects; and an unstructured covariance matrix structure for repeated intraindividual measurements. Adjusted mean absolute change in FAZ area from baseline was yielded from the mixed model with repeated measures analyses. In addition, descriptive statistics were provided for each end point.

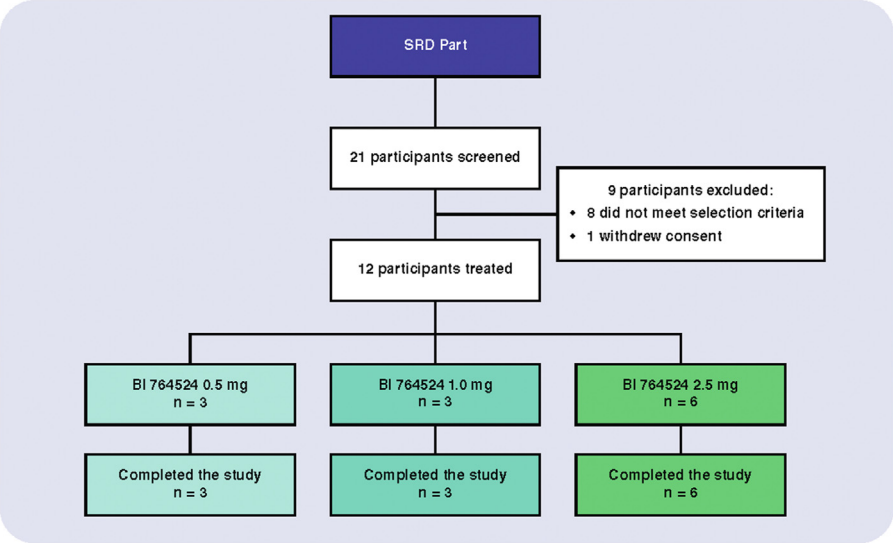


Figure 2. CONSORT flow diagram of patient disposition in the SRD part of the HORNBILL study. CONSORT = Consolidated Standards of Reporting Trials; SRD = single rising dose.

Results

SRD Part

Baseline Demographics and Characteristics. Twelve participants (female, n = 6 [50%]; White, n = 9 [75%]) were enrolled into 3 different consecutive dose groups after safety and tolerability were confirmed for each dose group by the safety monitoring committee (Fig 2 and Table 2). The mean (standard deviation [SD]) time since reported onset of DMI was 4.8 (4.6) years overall. At baseline, the mean (SD) age of the participant was 61.8 (8.3) years, and the mean (SD) hemoglobin A1c was 8.3 (1.3)%. The baseline study eye mean (SD) BCVA was 30.8 (21.7) letters.

Safety. No dose-limiting events were reported. One of the 6 participants in the BI 764524 2.5-mg group experienced 2 SAEs that were assessed by the investigator as not drug-related (peripheral vascular disorder and localized infection; Table 3). Four of the 12 participants (BI 764524 0.5 mg, n = 2 [67%]; BI 764524 2.5 mg, n = 2 [33%]) had ocular AEs in the study eye. None of the reported AEs was considered drug-related. The highest tested dose (BI 764524 2.5 mg) was selected for investigation in the MD part. Early pharmacodynamic end points for the SRD part of the study are reported in Table S4 (available at www.ophthalmologyscience.org).

MD Part

Baseline Demographics and Characteristics. Thirty-one participants (female, n = 15 [48%]; White, n = 26 [84%]) were randomized 2:1 into the BI 764524 or sham arm (Fig 3). The overall mean (SD) age of the participant was 59.5 (11.4) years, and the mean (SD) time since reported onset of proliferative DR was 1.9 (2.4) years (Table 5). Baseline mean (SD) study eye BCVA was 65.2 (18.6) letters. Although baseline characteristics between groups were generally well balanced, there was an imbalance in the mean FAZ area between treated and sham cohorts (0.72 vs. 0.51 mm^2). The primary data analysis was adjusted to account for the imbalance in baseline FAZ area.

Table 2. Participant Baseline Demographics and Characteristics in the SRD Part (Treated Set)

Parameter	BI 764524 0.5 mg (n = 3)	BI 764524 1.0 mg (n = 3)	BI 764524 2.5 mg (n = 6)
Sex, n (%)			
Female	2 (67)	2 (67)	2 (33)
Male	1 (33)	1 (33)	4 (67)
Race, n (%)			
White	1 (33)	2 (67)	6 (100)
Asian	1 (33)	1 (33)	0
Black/African American	1 (33)	0	0
Age, mean (SD), yrs	64.7 (15.0)	61.0 (7.0)	60.8 (6.0)
Time since DMI onset, mean (SD), yrs	2.7 (4.6)	4.6 (2.0)	5.9 (5.6)
BMI, mean (SD), kg/m^2	36.3 (8.9)	27.8 (4.6)	28.3 (4.6)
HbA1c, mean (SD), %	8.1 (1.1)	8.0 (1.9)	8.5 (1.2)
BCVA, mean (SD), letters			
Study eye	8.3 (14.4)	52.7 (10.3)	31.0 (17.9)
Fellow eye	40.3 (10.5)	72.3 (18.6)	66.7 (22.2)
IOP, mean (SD), mmHg			
Study eye	16.0 (3.6)	18.0 (2.6)	15.2 (2.6)
Fellow eye	15.7 (4.0)	15.3 (4.2)	15.3 (4.0)
FAZ area (FTR), mean (SD), mm^2	NR*	0.88 (0.25)	0.49 (0.45)
CST, mean (SD), µm	218.7 (150.8)	216.7 (47.9)	270.0 (101.2)

BCVA = best-corrected visual acuity; BMI = body mass index; CST = central subfield thickness; DMI = diabetic macular ischemia; FAZ = foveal avascular zone; FTR = full-thickness retina; HbA1c = hemoglobin A1c; IOP = intraocular pressure; NR = not reported; OCTA = OCT angiography; SD = standard deviation; SRD = single rising dose.
*No evaluable data collected regarding FAZ parameters due to OCTA image quality.

Table 3. Summary of AEs Reported in the SRD Part (Treated Set)

AE, n (%)*	BI 764524 0.5 mg (n = 3)	BI 764524 1.0 mg (n = 3)	BI 764524 2.5 mg (n = 6)	Total (N = 12)
Any AE	3 (100)	1 (33)	5 (83)	9 (75)
Severe	0	0	1 (17)	1 (8)
Peripheral vascular disorder	0	0	1 (17)	1 (8)
Serious	0	0	1 (17)	1 (8)
Localized infection	0	0	1 (17)†	1 (8)
Peripheral vascular disorder	0	0	1 (17)‡	1 (8)
Ocular AE, in the study eye	2 (67)	0	2 (33)	4 (33)
Conjunctival hemorrhage	1 (33)	0	0	1 (8)
Ocular hyperemia	1 (33)	0	0	1 (8)
Vitreous detachment	0	0	1 (17)	1 (8)
Procedural pain	0	0	1 (17)	1 (8)
Increased intraocular pressure	1 (33)	0	0	1 (8)
Procedure-related	2 (67)	0	1 (17)	3 (25)

AE = adverse event; SRD = single rising dose.
This table reports the numbers of participants with an AE. Individual participants may have had >1 AE.
*Percentages were calculated using the total number of participants per treatment group as the denominator.
†Medically important.
‡Requiring/prolonging hospitalization.

Safety. Two of the 11 AEs reported in the BI 764524 arm were assessed as drug-related by the investigator: vitreous floaters (mild) and increased gamma-glutamyl transferase levels (moderate; Table 6). Two patients experienced severe AEs (n = 1 [10%] sham, n = 1 [5%] BI 764524). Five patients (n = 3 [30%] sham, n = 2 [10%] BI 764524) reported 6 SAEs. None of the severe AEs or SAEs were considered drug-related. Seven (23%) of the 31 participants had an ocular AE in the study eye (n = 4 [40%] sham, n = 3 [14%] BI 764524); 1 event of vitreous floaters in the BI 764524 arm was considered by the investigator to be drug-related (Table 6). None of the participants receiving BI 764524 discontinued the study or treatment due to an AE compared with 2 of the 10 participants (20%) who received sham injections who discontinued due to AEs (Table 6). There were no reports of intraocular inflammation or occlusive vasculitis.

Early Pharmacodynamics. To assess the early pharmacodynamic effects of BI 764524, longitudinal changes from baseline to week 22 in FAZ area (in the FTR, based on

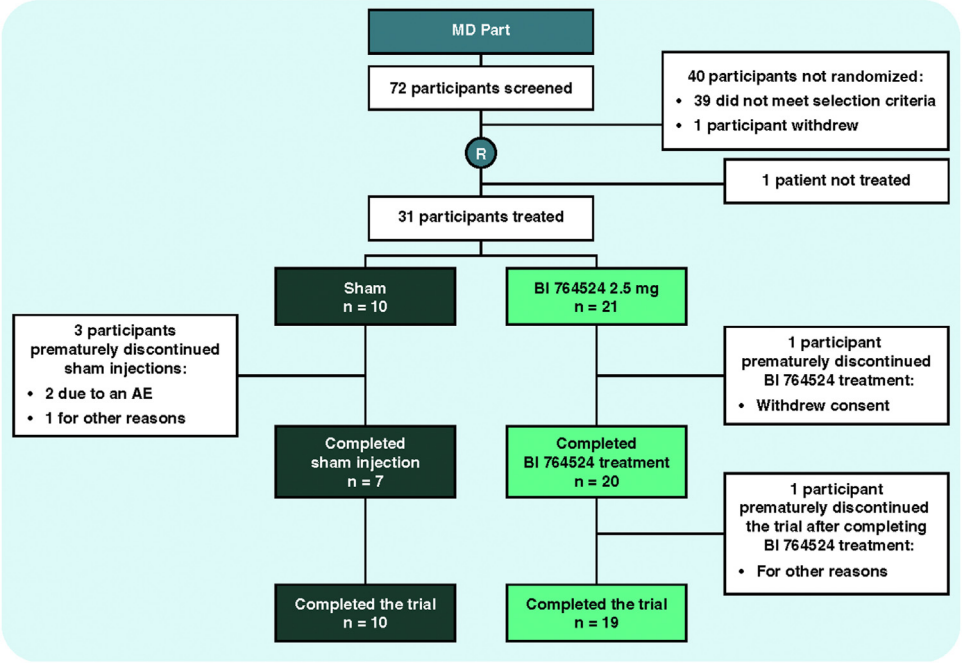


Figure 3. CONSORT flow diagram of patient disposition in the MD part of the HORNBILL study. AE = adverse event; CONSORT = Consolidated Standards of Reporting Trials; MD = multiple dose; R = randomization.

Table 5. Participant Baseline Demographics and Characteristics in the MD Part (Treated Set)

Parameter	Sham Injection (n = 10)	BI 764524 2.5 mg (n = 21)
Sex, n (%)		
Female	5 (50)	10 (48)
Male	5 (50)	11 (52)
Race, n (%)		
White	9 (90)	17 (81)
Black/African American	0	3 (14)
Asian	1 (10)	1 (5)
Age, mean (SD), yrs	60.0 (6.5)	59.3 (13.2)
Time since DMI onset, mean (SD), yrs	0.8 (0.9)	2.3 (2.7)
BMI, mean (SD), kg/m^2	32.8 (6.8)	29.4 (4.3)
HbA1c, mean (SD), %	8.0 (1.1)	8.6 (1.8)
BCVA, mean (SD), letters		
Study eye	69.5 (12.3)	63.1 (20.9)
Fellow eye	75.4 (7.7)	66.0 (21.0)
IOP, mean (SD), mmHg		
Study eye	15.6 (3.4)	16.0 (3.5)
Fellow eye	15.1 (3.0)	16.0 (3.1)
FAZ area (FTR), mean (SD), mm^2	0.51 (0.18)	0.72 (0.38)
CST, mean (SD), µm	262.0 (21.6)	246.6 (40.1)

BCVA = best-corrected visual acuity; BMI = body mass index; CST = central subfield thickness; DMI = diabetic macular ischemia; FAZ = foveal avascular zone; FTR = full-thickness retina; HbA1c = hemoglobin A1c; IOP = intraocular pressure; MD = multiple dose; SD = standard deviation.

Table 6. Summary of AEs Reported in the MD Part (Treated Set)

AE, n (%)*	Sham Injection (n = 10)	BI 764524 2.5 mg (n = 21)	Total (N = 31)
Any AE	8 (80)	11 (52)	19 (61)
Severe	1 (10)	1 (5)	2 (6)
Chronic kidney disease	1 (10)	0	1 (3)
Embolism	0	1 (5)	1 (3)
Serious	3 (30)	2 (10)	5 (16)
Vitreous hemorrhage	2 (20)†	0	2 (6)
Injection-site inflammation	0	1 (5)†	1 (3)
Streptococcal sepsis	0	1 (5)‡	1 (3)
Embolism	0	1 (5)‡	1 (3)
Chronic kidney disease	1 (10)‡	0	1 (3)
Ocular AE, in the study eye	4 (40)	3 (14)	7 (23)
Vitreous floaters	0	2 (10)	2 (6)
Vitreous hemorrhage	2 (20)	0	2 (6)
Visual impairment	0	1 (5)	1 (3)
Diabetic keratopathy	0	1 (5)	1 (3)
Conjunctival hemorrhage	1 (10)	0	1 (3)
Episcleritis	1 (10)	0	1 (3)
Iris neovascularization	1 (10)	0	1 (3)
Photopsia	1 (10)	0	1 (3)
Procedure-related	1 (10)	1 (5)	2 (6)
Investigator-defined, treatment-related	0	2 (10)	2 (6)
Vitreous floaters	0	1 (5)	1 (3)
Increased GGT	0	1 (5)	1 (3)
Leading to treatment discontinuation	2 (20)	0	2 (6)

AE = adverse event; GGT = gamma-glutamyl transferase; MD = multiple dose.
This table reports the numbers of participants with an AE. Individual participants may have had >1 AE.
*Percentages were calculated using the total number of participants per treatment group as the denominator.
†Medically important.
‡Requiring/prolonging hospitalization.

OCTA), CST (based on OCT), and BCVA were measured and analyzed. Variable OCTA image quality led to varying number of evaluable OCTA images at different time points. The trial was not powered to detect statistically significant changes in FAZ area.

At baseline, the mean FAZ area was 0.719 mm^2 in the BI 764524 2.5-mg group and 0.512 mm^2 in the sham group. At week 12 (4 weeks after the final injection) and at week 22 (14 weeks after the final injection), the mean FAZ area was 0.769 mm^2 and 0.769 mm^2 in the BI 764524 group and 0.546 mm^2 and 0.553 mm^2 in the sham group. The adjusted mean absolute change in FAZ area from baseline at week 12 and week 22 was −0.004 mm^2 and -0.001 mm^2 in the BI 764524 2.5-mg group compared with +0.019 mm^2 and +0.010 mm^2 in the sham group (difference between groups at week 12: −0.023 mm^2, 1-sided $P = 0.07$ and at week 22: difference: −0.011 mm^2, 1-sided $P = 0.23$) (Table 7, Fig 4, and Fig S5, available at www.ophthalmologyscience.org). An additional analysis of the percent change from baseline in FAZ area showed a week-12 change of −1.5% in the BI 764524 2.5-mg group and +3.9% in the sham group (difference: −5.4%, Table 7). At week 22, percent changes were −0.3% in the BI 764524 2.5-mg group and +2.8% in the sham group (difference: −3.2%). A summary of early pharmacodynamic results is provided in Table S8 (available at www.ophthalmologyscience.org).

Adjusted mean change from baseline in BCVA at week 12 was +1.5 letters in the BI 764523 group (n = 17) and +4.8 letters in the sham group (n = 10) (difference: −3.3 letters; $P = 0.2880$) (Fig 6; Tables S9, S10, available at www.ophthalmologyscience.org). At week 22, adjusted mean change was +3.1 letters in the BI 764523 group (n = 19) and +5.8 letters in the sham group (n = 10) (difference: −2.8 letters; $P = 0.3405$). Adjusted mean change from baseline in CST at week 12 was +3.7 µm in the BI 764523 group (n = 17) and −3.9 µm in the sham group (n = 9) (difference: 7.7 µm; $P = 0.0874$). At week 22, adjusted mean change was +3.4 µm in the BI 764523 group (n = 19) and −3.4 µm in the sham group (n = 9) (difference: 6.8 µm; $P = 0.0808$).

Discussion

HORNBILL was a first-in-human phase I/IIa study investigating the safety, tolerability, and early pharmacodynamic response to IVT administration of BI 764524 in patients with DMI secondary to DR. To our knowledge, it was also the first clinical trial to investigate a pharmacologic agent of any target for DMI. Demonstrating IVT administration of BI

Table 7. Adjusted Mean Change in FAZ Area (FTR) of the Study Eye Over Time in the MD Part of the Study

	Sham Injection			BI 764524 2.5 mg			Adjusted Mean Difference		
Week	*n*	*Adjusted Mean Absolute Change (mm^2)**	*Adjusted Mean Percent Change†*	*n*	*Adjusted Mean Absolute Change (mm^2)**	*Adjusted Mean Percent Change†*	*Absolute (mm^2)**	*Percent (%)†*	**P Value (One-Sided)***
4	5	−0.001	0.7	17	−0.004	−1.2	−0.003	−1.9	0.43
8	6	0.013	3.2	15	−0.008	−1.6	−0.021	−4.8	0.09
12	7	0.019	3.9	16	−0.004	−1.5	−0.023	−5.4	0.07
16	7	0.008	2.5	15	−0.005	−1.7	−0.013	−4.2	0.16
22	7	0.010	2.8	16	−0.001	−0.3	−0.011	−3.2	0.23

FAZ = foveal avascular zone; FTR = full-thickness retina; MD = multiple dose; MMRM = mixed model for repeated measures.
*Data based on an MMRM (applied to absolute changes in FAZ area from baseline) that included baseline value as a continuous covariate, visit and treatment as fixed categorical effects, and visit by treatment and visit by baseline interactions, with an unstructured covariance matrix structure for repeated measurements within individual participants.
†Data based on an MMRM (applied to percentage changes in FAZ area from baseline) that included baseline value as a continuous covariate, visit and treatment as fixed categorical effects, and visit by treatment and visit by baseline interactions, with unstructured covariance matrix structure for repeated measurements within individual participants.

764524 to be well tolerated after both single-dose and MD administration up to the highest tested dose of 2.5 mg, HORNBILL met its primary safety end points. The frequency of ocular AEs in the study eye was comparable with other IVT drugs.[21] Of note, there were no reports of intraocular inflammation or occlusive vasculitis. None of the participants who received BI 764524 discontinued the treatment or trial due to drug-related AEs or experienced dose-limiting AEs. In the MD part, 2 of the 20 patients receiving BI 764524 experienced AEs that were deemed by the investigator to be drug-related, warranting further investigation in larger clinical trials of BI 764524. At this early stage of development, it is not possible to conclude whether these events were caused by the drug or by other factors, such as worsening of underlying diseases, and these events will be monitored in upcoming trials. The overall favorable safety profile of BI 764 524 in HORNBILL supports further investigation of this compound for IVT delivery.

In addition to the assessment of safety and tolerability, this study assessed early pharmacodynamic effects using changes in FAZ measurements, BCVA, and CST to inform further investigation of BI 764524 in diseases characterized by RNP. Several factors informed the selection of patients with DMI as the target trial population, including the high unmet need for therapies that can restore the central vasculature in patients with DR and the recent technologic advancements in the field of OCTA imaging.[7,8]

Reported changes in functional parameters in patients with DMI over 1 year are minimal.[22] As such, significant changes in BCVA were not expected to be seen over 22 weeks of follow-up in the HORNBILL trial. Consistently, there were no relevant differences observed between sham and BI 764524 in this trial. Similarly, observed changes in CST are within the expectations for an early-stage trial and did not show obvious differences between sham and BI 764524 treatment.

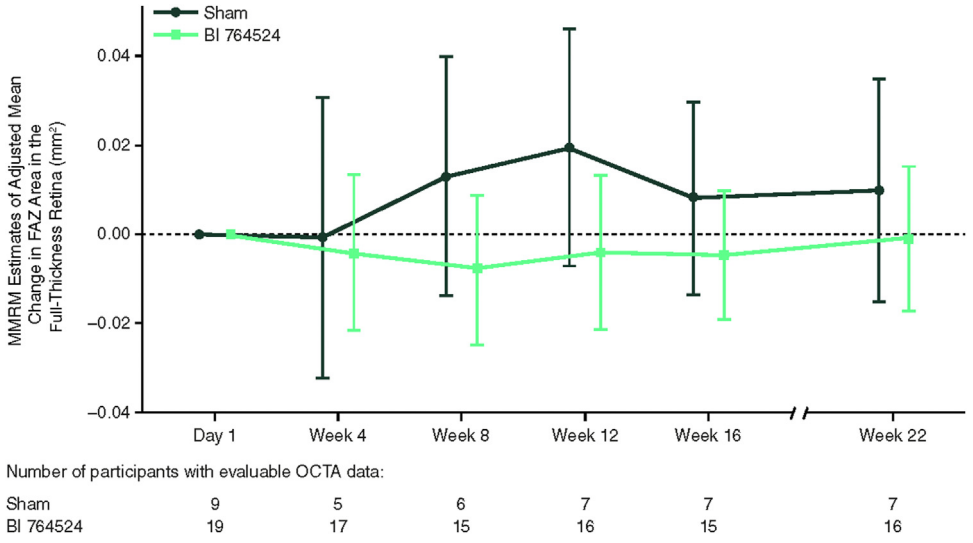


Figure 4. Adjusted mean (95% CI) changes in FAZ area (mm^2; FTR) for patients who received BI 764524 (2.5 mg) or sham injections (MMRM analysis). Error bars are showing 95% CIs. CI = confidence interval; FAZ = foveal avascular zone; FTR = full-thickness retina; MMRM = mixed model for repeated measures; OCTA = OCT angiography.

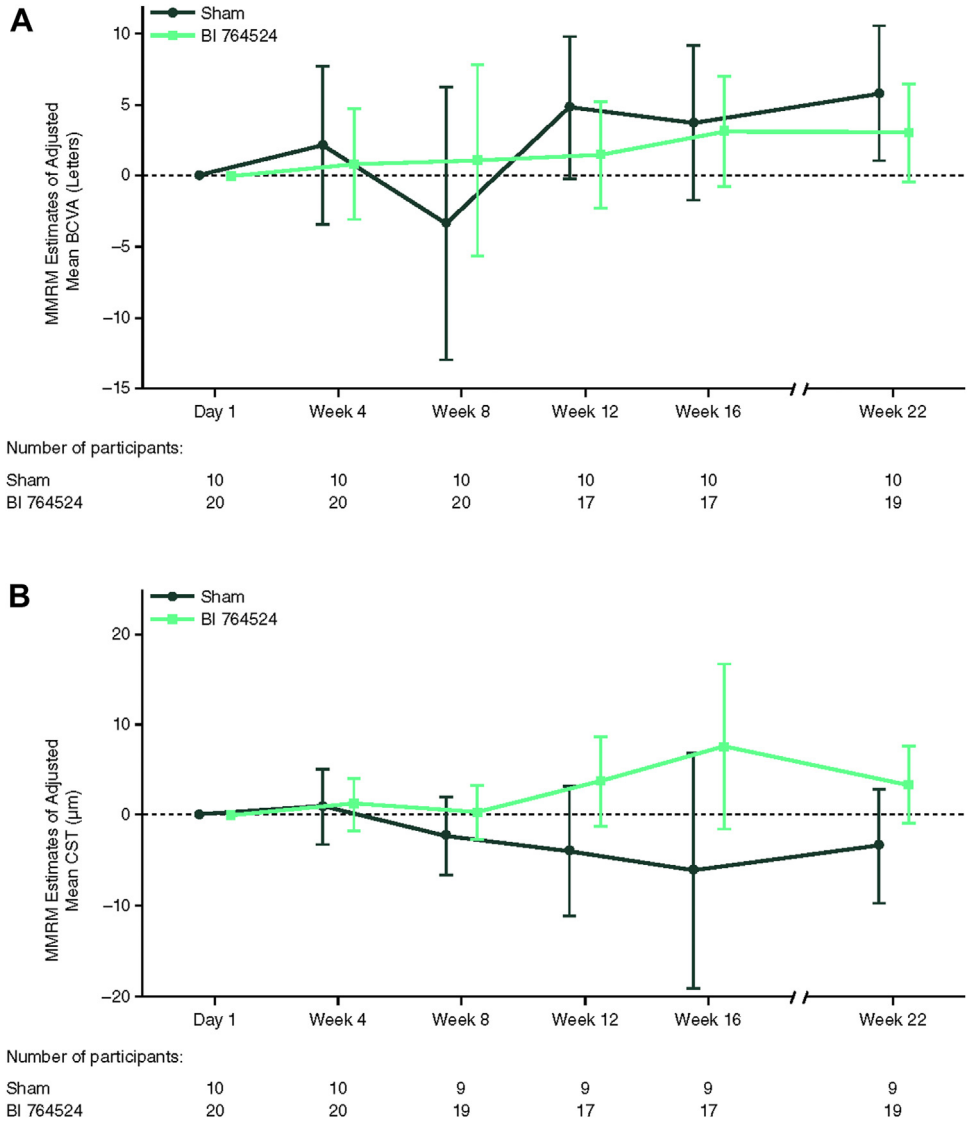


Figure 6. Adjusted mean (95% CI) change from baseline to week 22 in **(A)** BCVA (letters) and **(B)** CST for participants who received BI 764524 (2.5 mg) or sham injections (MMRM analysis). BCVA = best corrected visual acuity; CI = confidence interval; CST = central subfield thickness; MMRM = mixed model for repeated measures.

Before the trial, it was hypothesized that OCTA imaging may allow for the detection of microvascular changes indicative of disease progression before functional changes become detectable,[22] and it was hypothesized that increases in FAZ area may be detected on OCTA images taken throughout the trial. Therefore, in the HORNBILL study, change in FAZ area over time was selected as an additional exploratory pharmacodynamic parameter. Consensus on the optimal approach for FAZ area measurement using OCTA has not yet been reached;[8] there is considerable disagreement with regard to the optimal retinal layer segmentation and methods of FAZ area measurement. Considering these views, before the final analysis of HORNBILL, masked trial data were analyzed in both the FTR and the superficial vascular complex with different scan sizes. Foveal avascular zone measurements derived from 6 × 6-mm scans of the superficial vascular complex showed considerably higher variability than those derived from 3 × 3-mm FTR scans. This increase in variability may be partly attributed to differences in resolution. To avoid suboptimal resolution skewing the data, an FAZ area derived from 3 × 3-mm OCTA scans of the FTR was preselected for the final analysis. This approach aligns with studies indicating that smaller 3 × 3-mm OCTA scans have greater resolution and can delineate FAZ with higher accuracy than 6 × 6-mm OCTA scans.[23–25]

Among sham-treated participants, a trend toward increasing FAZ size was seen. This was slower than initially anticipated but within the range of recently published data.[22,23] In the BI 764524 2.5-mg dose group, the mean FAZ size showed an increase, whereas the adjusted mean absolute change in FAZ area at week 12

was −0.004 mm^2. This difference can be explained by the varying number of participants with available data for analysis at the different timepoints and also highlights the challenges of analysis of datasets with small sample sizes. Overall, due to the slow progression of the sham arm, the limited separation between both groups, and the considerable variability in individual FAZ changes over time in the HORNBILL study, it is difficult to draw robust conclusions from these exploratory observations. Furthermore, the repeatability and consistency of scan measurements captured within a short period of time remain unclear. Of note, the current phase I/IIa study was not powered, and the duration was too short to demonstrate a statistically significant therapeutic effect of BI 764524 in DMI.

These results highlight that future trials in DMI and RNP will require a longer duration and larger sample size to achieve sufficient statistical power for efficacy assessment. Further research will also be needed to establish FAZ as a reliable imaging biomarker for RNP and to understand the association between RNP and functional DMI outcomes. Identification of appropriate patient populations with dynamic progression of RNP might help to select patients who would benefit the most from a treatment capable of reducing RNP. However, the favorable safety profile of BI 764524 provides a good foundation for further investigations of BI 764524 in patients with DR and RNP.

Conclusion

Overall, IVT administration of BI 764524 to patients with DR was well tolerated in this first-in-human phase I/IIa study. The results of the HORNBILL study support further investigation of BI 764524 in patients with DR and RNP. This is currently ongoing in the phase II CRIMSON study.

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Data Sharing Statement. To ensure independent interpretation of clinical study results and enable authors to fulfill their role and obligations under the ICMJE criteria, Boehringer Ingelheim grants all external authors access to relevant clinical study data. In adhe rence with the Boehringer Ingelheim Policy on Transparency and Publication of Clinical Study Data, scientific and medical researchers can request access to clinical study data, typically 1 year after the approval has been granted by major Regulatory Authorities or after termination of the development program. Researchers should use the https://vivli.org/ link to request access to study data and visit https://www.mystudywindow.com/msw/datasharing for further information.

Footnotes and Disclosures

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A lay summary of the clinical trial results was developed by the trial sponsor and is available here: https://www.mystudywindow.com/trial/completed/241496/1436-0001. A plain language summary of the clinical trial results was also developed and is available here: [https://doi.org/10.60915/Boehringer-figshare.29220842.v1].

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HUMAN SUBJECTS: Human subjects were included in this study. The HORNBILL study was conducted across 19 clinical centers in the United Kingdom and the United States. Institutional Review Board/Ethics Committee approval was obtained. The study was conducted in accordance with the principles of the Declaration of Helsinki and the International Conference on Harmonisation Good Clinical Practice guidelines. All study participants provided written informed consent before enrollment.

No animal subjects were used in this study.

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Abbreviations and Acronyms:
AE = adverse event; **BCVA** = best-corrected visual acuity; **CST** = central subfield thickness; **DMI** = diabetic macular ischemia; **DR** = diabetic retinopathy; **FAZ** = foveal avascular zone; **FTR** = full-thickness retina; **IVT** = intravitreal; **MD** = multiple dose; **OCTA** = OCT angiography; **RNP** = retinal nonperfusion; **SAE** = serious adverse event; **SD** = standard deviation; **Sema3A** = semaphorin 3A; **SRD** = single rising dose.

Keywords:
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